

Sieng Lyheng

Applied Math & Statistics — Data Science Student

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EDUCATION

Institute of Technology of Cambodia

Phnom Penh, Cambodia

B.Eng. Applied Mathematics & Statistics (Data Science)

Expected 2027

Year 3 GPA: 4.00/4.00

EXPERIENCE

Data Visualization for Library Resources and Attendance using Python

Aug 2024 – Sep 2024

ITC Library, Informal Internship

Phnom Penh, Cambodia

- Analyzed borrowing and room booking data from multiple departments using Python and Jupyter Notebook.
- Reported findings and insights to the Head of Library to support decision-making.

PROJECTS

Sao Somleng Game - Khmer Voice Game

2026

Python, FastAPI, PyTorch, Wav2Vec2, React

- Developed a voice-controlled educational game to help children learn Khmer characters by speaking the character shown on the wall.
- Fine-tuned Wav2Vec2 audio classification models and used a two-model ensemble for Khmer character voice prediction.
- Added continuous play mode and an analysis page to track practice performance.

HIV Molecules Inhibitors Classification

2026

Python, GNN, Graph Transformer Network, PyTorch Geometric, RDKit, MoleculeNet

- Developed a graph-based deep learning project to classify molecules with potential HIV inhibitory activity.
- Supported early-stage drug discovery through molecular structure analysis.

Alphabet Recognition

2026

Python, TensorFlow/Keras, CNN, NumPy, Matplotlib

- Built a deep learning model to recognize handwritten English alphabets from A–Z.
- Used CNN and AlexNet-inspired ideas for educational handwriting recognition.

Automatic License Plate Recognition in Cambodia (ALPRC)

2025

Python, YOLOv8, OCR, PaddleOCR, EasyOCR, Roboflow, Flask

- Developed a system for detecting and recognizing Cambodian vehicle license plates.
- Applied YOLOv8 and OCR techniques for plate detection and text extraction.

Library Management System

2025

Django, Python, MySQL, HTML, CSS, JavaScript

- Developed a web-based system for managing books, users, borrowing, returns, fines, and reports.
- Improved library work by changing manual tasks into a digital system.

TECHNICAL SKILLS

Languages: Python, R, SQL, Java, C, C++

Data Science/ML: Pandas, Scikit-learn, TensorFlow/Keras, PySpark, Data Analysis, Machine Learning, Deep Learning

Visualization: Power BI, Microsoft Excel, Matplotlib, Jupyter Notebook

Databases: MySQL, PostgreSQL, MongoDB

Tools: GitHub, VS Code, MySQL Workbench, Code::Blocks

LANGUAGES

Khmer (Native), English (Intermediate)